

COS 730 – Assignment 2

From Behavioural Models to Optimised Implementation

Due date: 14th May 2026 @ 11:00 am

Overview

In real-world software engineering, initial designs are often **functionally correct but suboptimal**. Engineers must be able to:

- Translate behavioural models into working implementations
- Critically evaluate design inefficiencies
- Optimise object interactions and decision logic
- Demonstrate measurable improvements

In this assignment, you will be provided with a **sequence diagram representing a system design**. You are required to:

1. Implement the system as specified
2. Analyse inefficiencies in the design
3. Optimise the interaction model and decision logic
4. Re-implement the improved design
5. Empirically demonstrate performance and design improvements

You will be given a **baseline sequence diagram** representing a subsystem of a:

Intelligent Submission and Review System

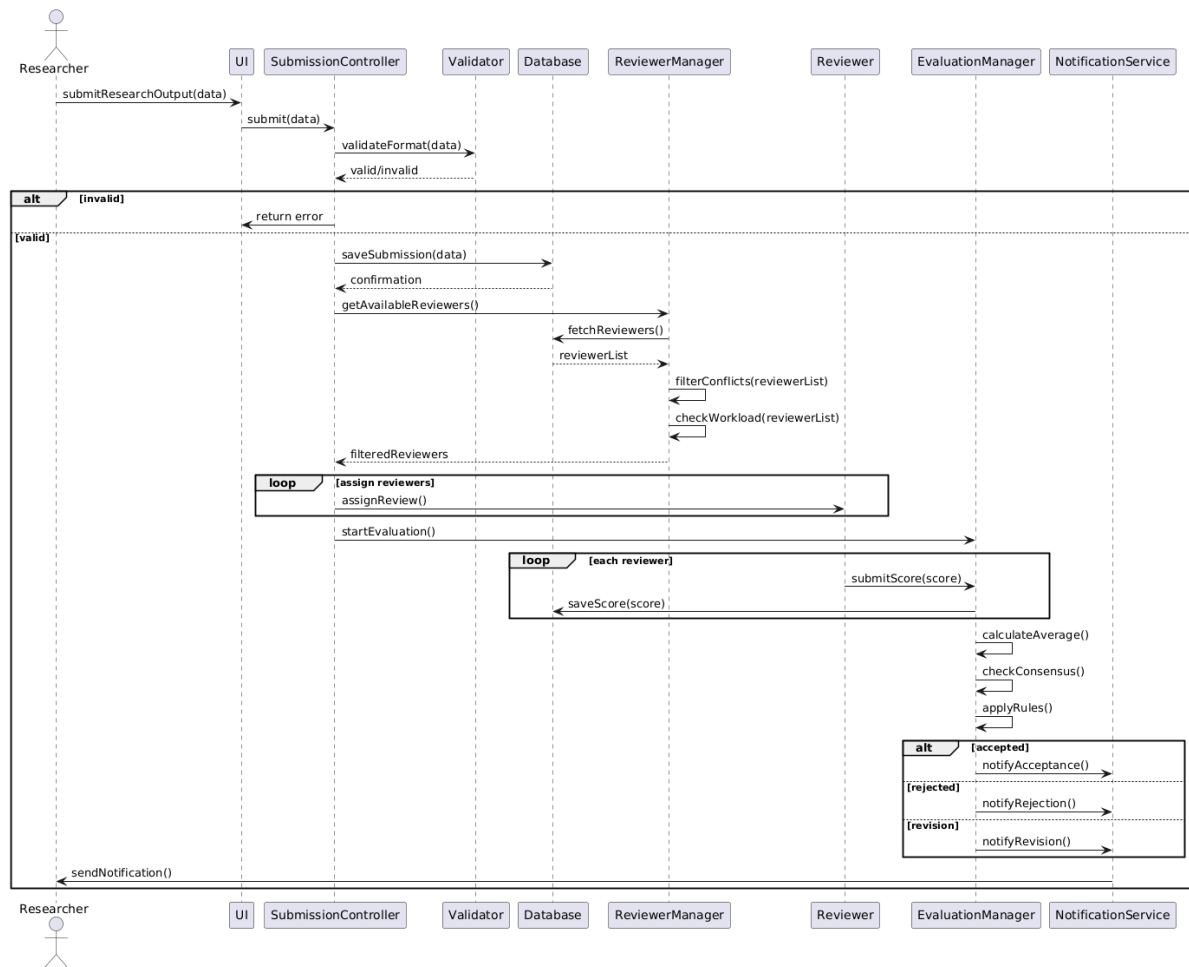
The sequence diagram models the process of:

- Submitting an artefact
- Validating submission rules
- Assigning reviewers
- Performing evaluation checks
- Producing a final outcome

The provided design intentionally includes:

- Redundant interactions
- Poor responsibility allocation
- Inefficient decision handling
- Tight coupling between components

Tasks



Task 1: Baseline Implementation (Correctness Phase) [15 marks]

You must:

1. Implement the system **exactly as specified** in the provided sequence diagram.
2. Ensure:
 - All interactions are preserved
 - Object responsibilities match the diagram
 - No optimisations are introduced at this stage

Requirements:

- Use an object-oriented language (e.g., Java, Python, C#)
- Maintain traceability between diagram elements and code

Task 2: Design Analysis - [10 marks]

Critically evaluate the given design.

Identify and explain:

- Redundant message calls
- Violations of good responsibility assignment
- High coupling / low cohesion areas
- Inefficient control flow
- Poor handling of decision logic

You must explicitly relate issues to:

- Responsibility assignment principles
- Behavioural modelling quality

Task 3: Decision Modelling Using a Decision Table – [15 marks]

Extract all decision logic from the system and represent it using a:

Decision Table

Your table must:

- Clearly define conditions and actions
- Eliminate ambiguity in decision-making
- Replace scattered conditional logic in the sequence diagram

You must also:

- Explain how the decision table improves clarity and maintainability
- Map decision table entries back to system behaviour

Task 4: Sequence Diagram Optimisation – [10 marks]

Redesign the sequence diagram to:

- Reduce unnecessary interactions
- Improve responsibility allocation
- Decouple system components
- Centralise or properly distribute decision logic

Your optimised design must demonstrate:

- Improved cohesion

- Reduced coupling
- Cleaner control flow
- Better separation of concerns

Task 5: Optimised Implementation – [20 marks]

Implement the **improved sequence diagram**.

You must:

- Refactor or redesign your initial implementation
- Align code with the optimised model
- Ensure functional equivalence with the original system

Task 6: Empirical Evaluation and Comparison – [30 marks]

You must compare the **baseline vs optimised system** using:

Metrics (at minimum):

- Number of method calls / interactions
- Execution time (benchmarking repeated runs)
- Code complexity (e.g., method size, class responsibilities)
- Maintainability indicators (qualitative + quantitative)

Analysis must include:

- Before vs after comparison
- Justification of improvements
- Trade-offs introduced by optimisation

Deliverables

You must submit:

Technical PDF Report (15 – 20 pages)

The report must include:

- Design analysis
- Identified issues
- Decision table construction
- Optimisation rationale
- Implementation discussion / screenshots
- Empirical evaluation and results

ABSOLUTELY NO LATE SUBMISSIONS!!!

TOTAL: 100